Quiz 4 – Uncertainty & NNs

Instructions: This is a multiple-choice question section. For each question, choose **only one** correct answer.

*** Enter your name and ID**

*** (1)Name**

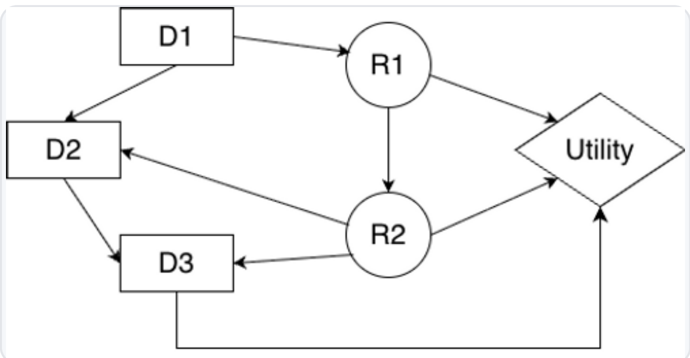
请输入

*** (2)Student ID**

请输入

1. [单选] **Decision Networks** (0.5分)

Given the decision network below, which is a order of variable elimination?



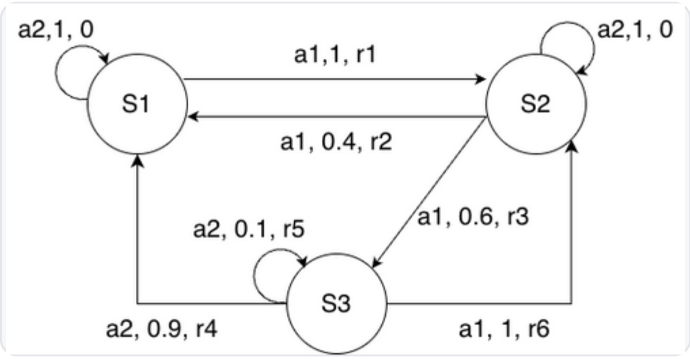
Since R2 is a parent of a decision node,
It shouldn't be eliminated first.

R1 is not the a parent of any decision node, it should be eliminated first. After that, the decision node D3 is the last decision node (D1->D2->D3), thus, D3 will be eliminated after R1.

- (selected) R1, D3, D2, R2, D1
- R2, D3, D2, R1, D1
- R1, D1, D2, D3
- R2, R1, D3, D2, D1

2. [单选] **Markov Decision Process (MDP)** (0.5分)

Consider the following transition graph of an MDP. Suppose there are two possible actions a1 and a2 for each state and the transitions are labeled with the tuple (action, probability of transition, reward). Let S1 be the start state. Which of the following sequence is NOT a possible run of this MDP?



- ◯ S1, a2, 0, S2, a1, r1, S2, a1, r2, S1
- ◯ S1, a1, r1, S2, a1, r3, S3, a2, r5, S3
- ◯ S1, a1, r1, S2, a1, r3, S3, a2, r4, S1
- ◉ S1, a2, 0, S2, a1, r1, S2, a1, r2, S3

The first and last option are not possible run of this MDP.
Selecting one of these two will get mark.

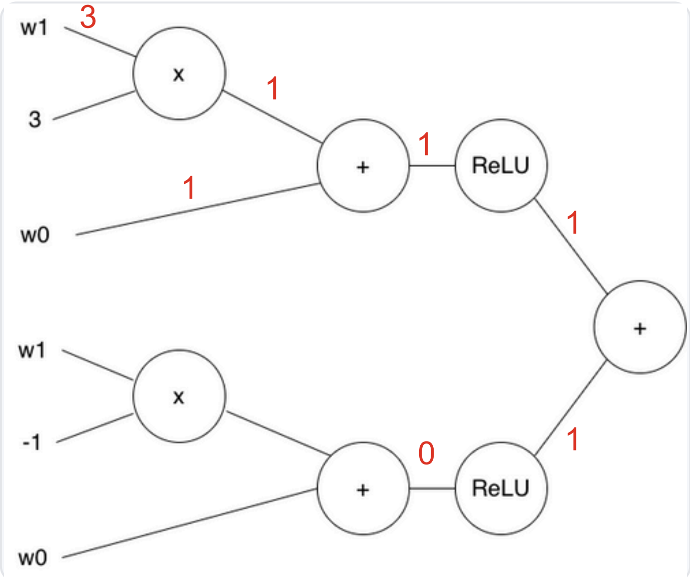
3. [单选] **Activation Functions** (0.5分)

Which of the following activation function will NOT saturate for large positive/negative inputs?

- ◯ Sigmoid
- ◯ Tanh
- ◯ ReLU
- ◉ LeakyReLU

4. [单选] **Backpropagation** (0.5分)

Provided the following computation graph, what is the gradient value at w0=1, w1=2?



Given w0=1, w1=2, the input to the bottom ReLU is -1 < 0. ReLU has zero gradient for its negative region. So, when performing BP, the bottom branch will have 0 contribution to the gradient of w0 and w1

- ○ [1, 2]
- ◉ [1, 3]
- ○ [2, 3]
- ○ [2, 2]

5. [单选] **Optimization** (0.5分)

Which of the following statements is FALSE for NN optimization?

- ○ SGD+Momentum handles saddle points by considering the velocity of the past steps.
- ○ SGD+Momentum is sensitive to the learning rate.
- ◉ RMSProp assigns lower learning rate for weights with smaller gradients
- ○ Adam can handle saddle points and adjust the learning rate during learning.

提 交